

BASICS OF ECONOMICS

MGT404

Government Interventions

Yale

Recap and Today's Plan

Last Lecture

- Market Equilibrium, consumer and producer surplus
- Elasticity of supply and demand
- Complements and substitutes

Today

- How do policy interventions in markets affect outcomes?
 - Tax Incidence
 - Externalities
 - Rationing, quotas, price controls, cap and trade.

Today's Plan

Government interventions:



1. Taxation

- **How taxes work**
- **Dead weight loss**
- **Who pays (elasticity matters)**

2. Application

- Soda tax article

3. Externalities

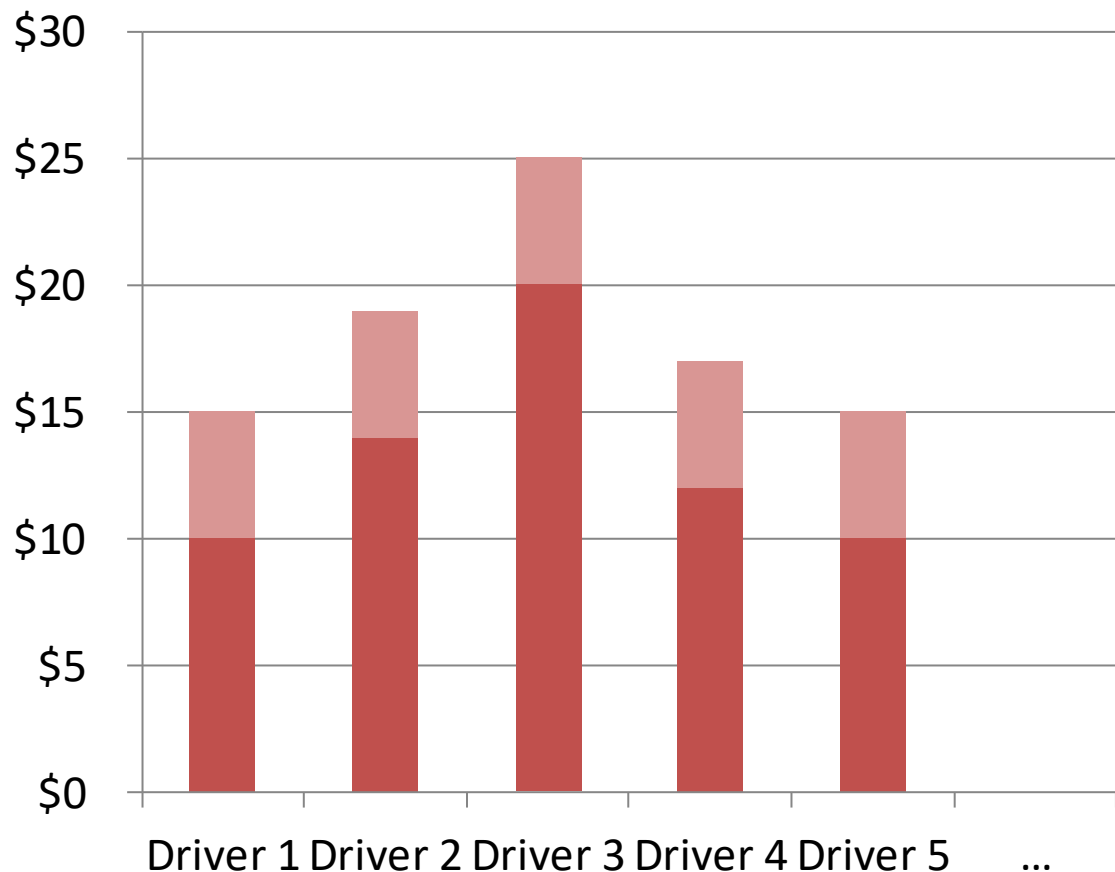
- Tax objectives
- Introduction to externalities
- “Corrective” taxation

4. Other policy tools

- Quotas/rationing
- Price floors
- Cap and Trade

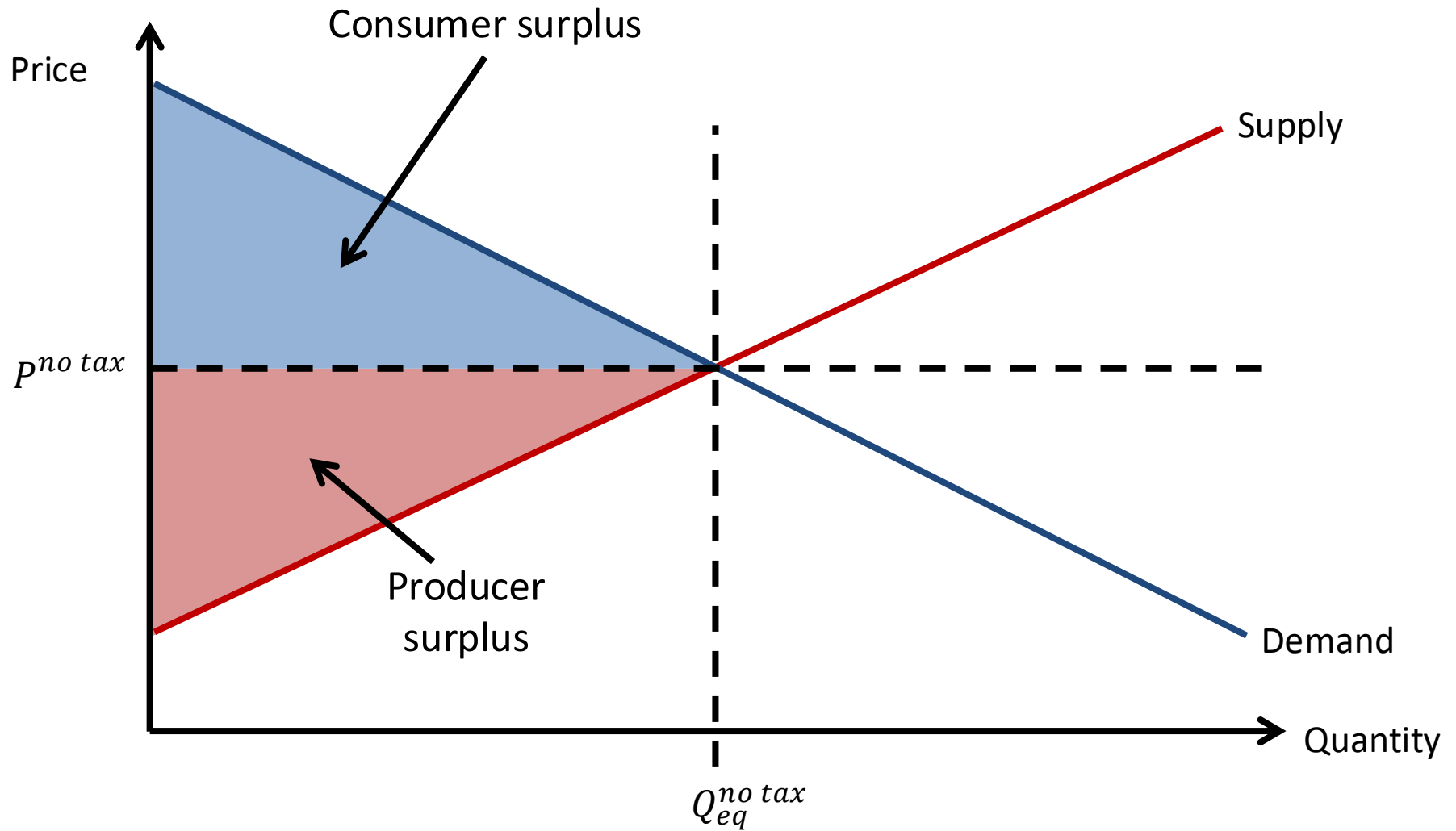
Effect of Introducing a Tax on Supply

Supply Reservation Prices: Uber driving

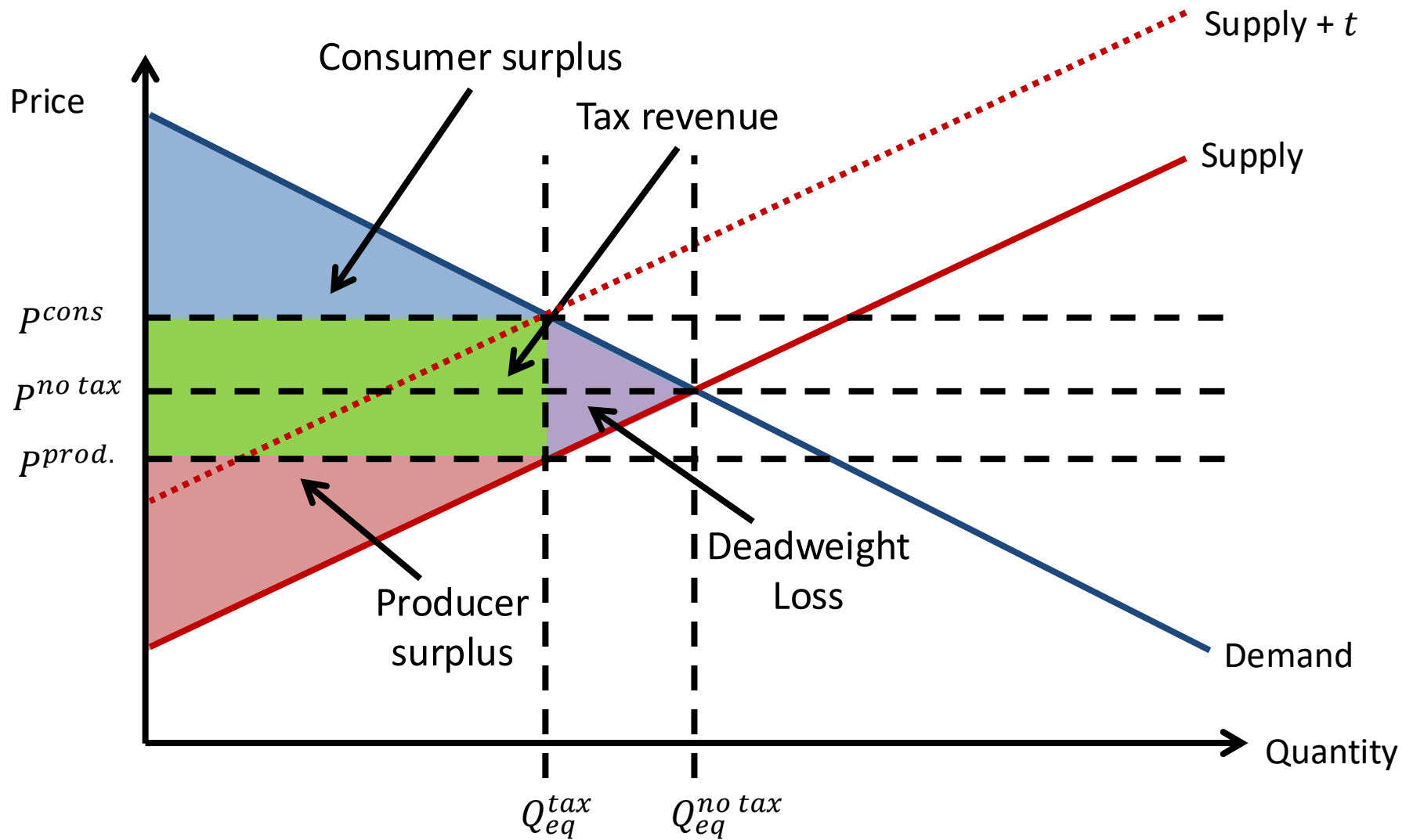


- What if drivers are required to pay a \$5 tax out of their earnings?
- Price must be \$5 higher before they will be willing to drive.
- Shifts supply curve **up** by amount equal to the tax.

Recall: Basic Market Equilibrium

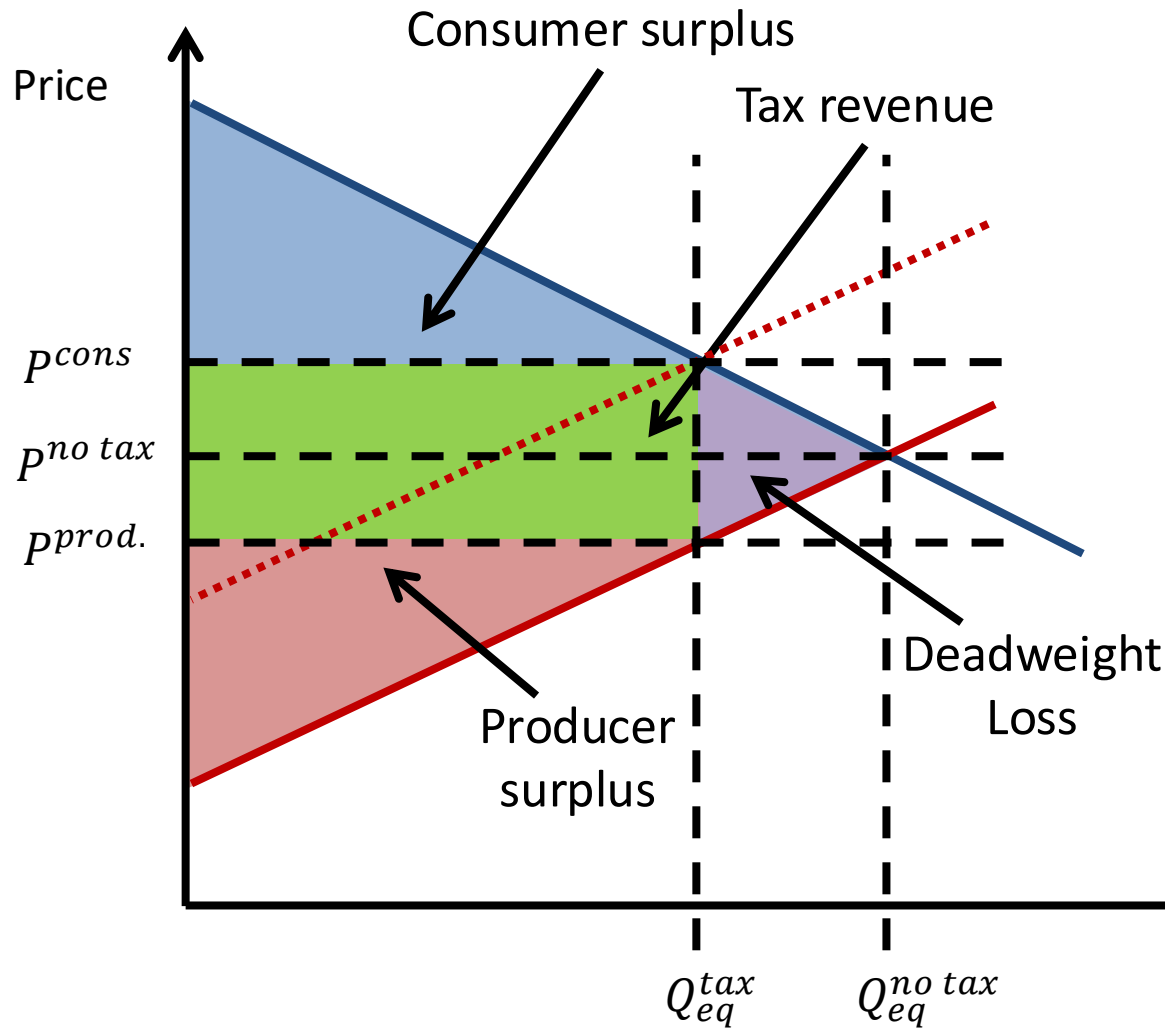


Introduce a Tax, t , Added to Price



Taxation creates a distortion in the market – this can decrease efficiency.

Introduce a Tax, t , Added to Price

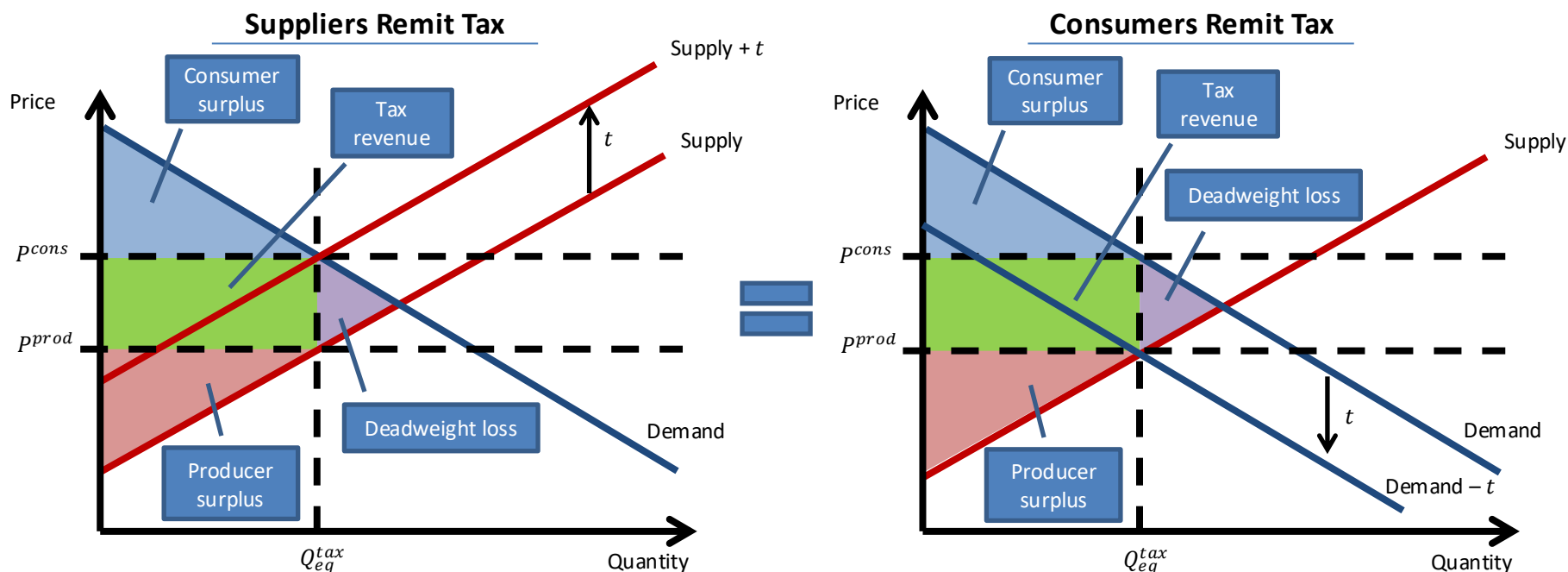


Three Important Concepts

1. A portion of the tax revenue comes out of the Consumer Surplus. Consumers *pay* this portion of the tax.
2. Similarly, producers *pay* the part of the tax revenue that comes from Producer Surplus.
3. The tax itself creates a **deadweight loss** – wasted potential surplus. Gains from trade are not being realized. This is inefficient.

Does it matter who remits the tax?

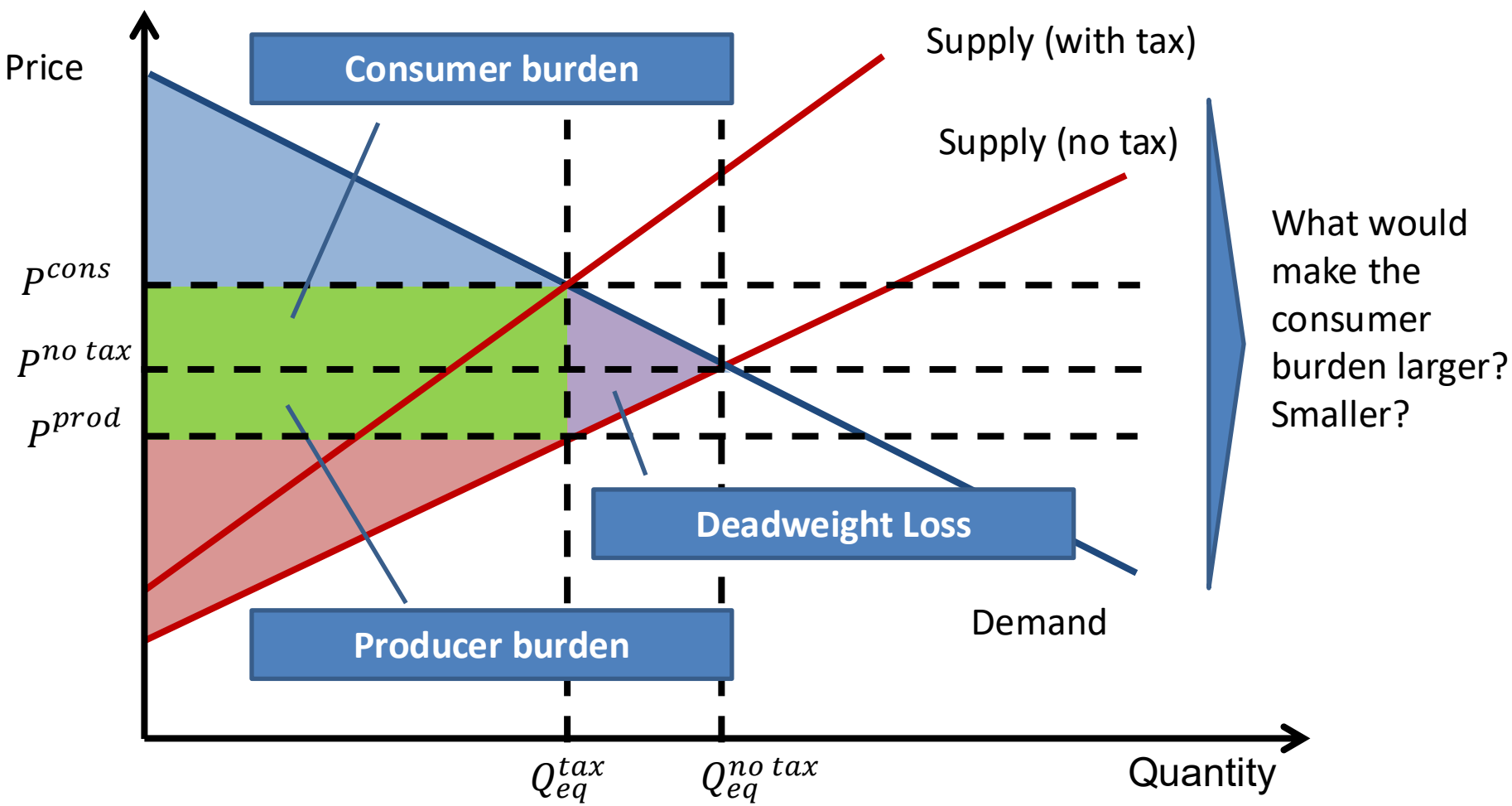
- Note: price consumers pay, P^{cons} , goes up, and price producers receive, P^{prod} goes down. Both pay a portion of the tax, effectively.
- This is true even if the tax is collected entirely from producers, since the *price adjusts*.



- In equilibrium, it doesn't matter whether the tax is levied on consumers or producers—the true burden of the tax will be the same.
- In class, we will always shift supply to keep it simple.

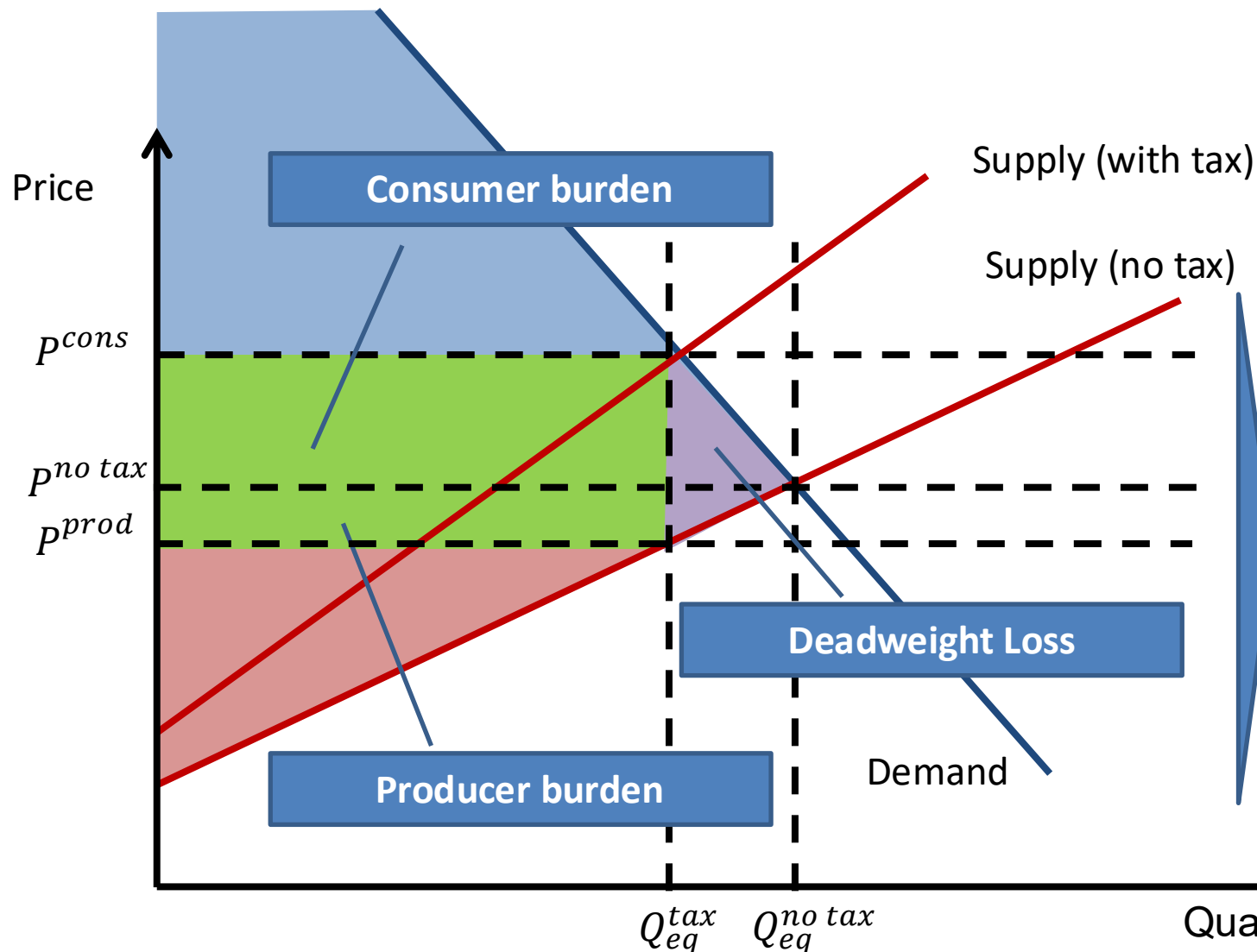
“Incidence” of Government Interventions: Who “Pays”?

Suppose there is a percentage tax imposed on manufacturers:



Elasticity Matters:

Suppose demand is more inelastic:

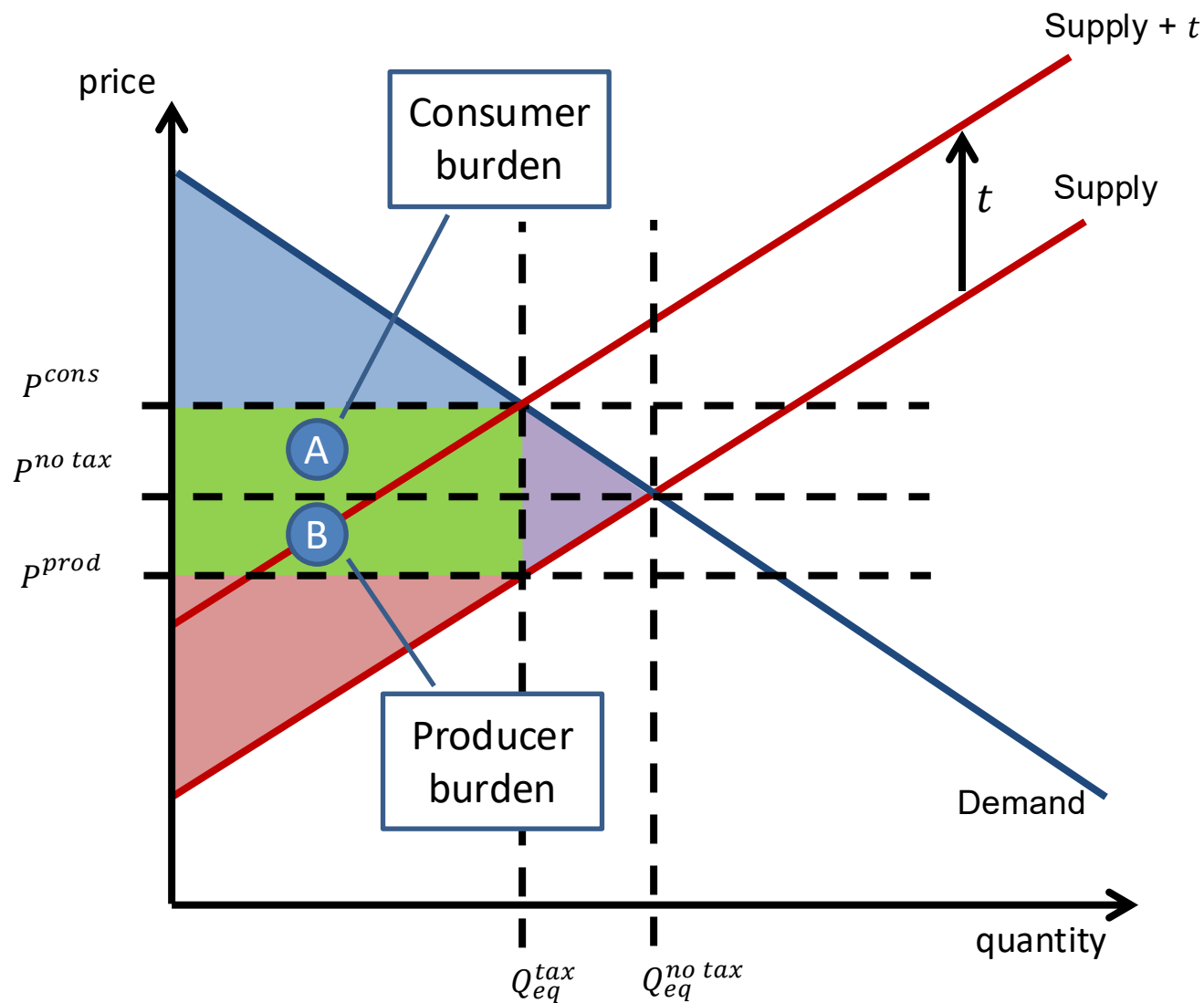


Here, the consumer burden is much larger than producer burden.

- Inelastic demand: consumers pay.
- Elastic demand: producers pay.

Note: inelastic side of the market always pays more. "Less nimble".

Tax burden formula



What fraction of the tax is paid by consumers?

We want to get: $\frac{A}{A+B}$

So we need:

$$\frac{(p^{cons} - p^{no\ tax}) * Q_{eq}^{tax}}{(p^{cons} - p^{prod}) * Q_{eq}^{tax}}$$

But notice that's just:

$$\frac{p^{cons} - p^{no\ tax}}{p^{cons} - p^{prod}}$$

Steps to calculating tax burden

Step 1

- Shift supply by the amount of the tax.
 - Per unit tax of t : $P_S^{tax} = P_S + t$
 - Percentage tax rate of t : $P_S^{tax} = (1 + t) \cdot P_S$

Step 2

- Calculate the new equilibrium quantity and prices (price paid by consumers and price received by producers).
 - Now set $P_D = P_S^{tax}$

Step 3

- Option A: compute areas of tax revenue, and determine area taken from consumer vs producer surplus.
- Option B: Use the price formula:

$$B_{cons} = \frac{P^{cons} - P^{no\ tax}}{P^{cons} - P^{prod}}, B_{prod} = \frac{P^{no\ tax} - P^{prod}}{P^{cons} - P^{prod}}$$

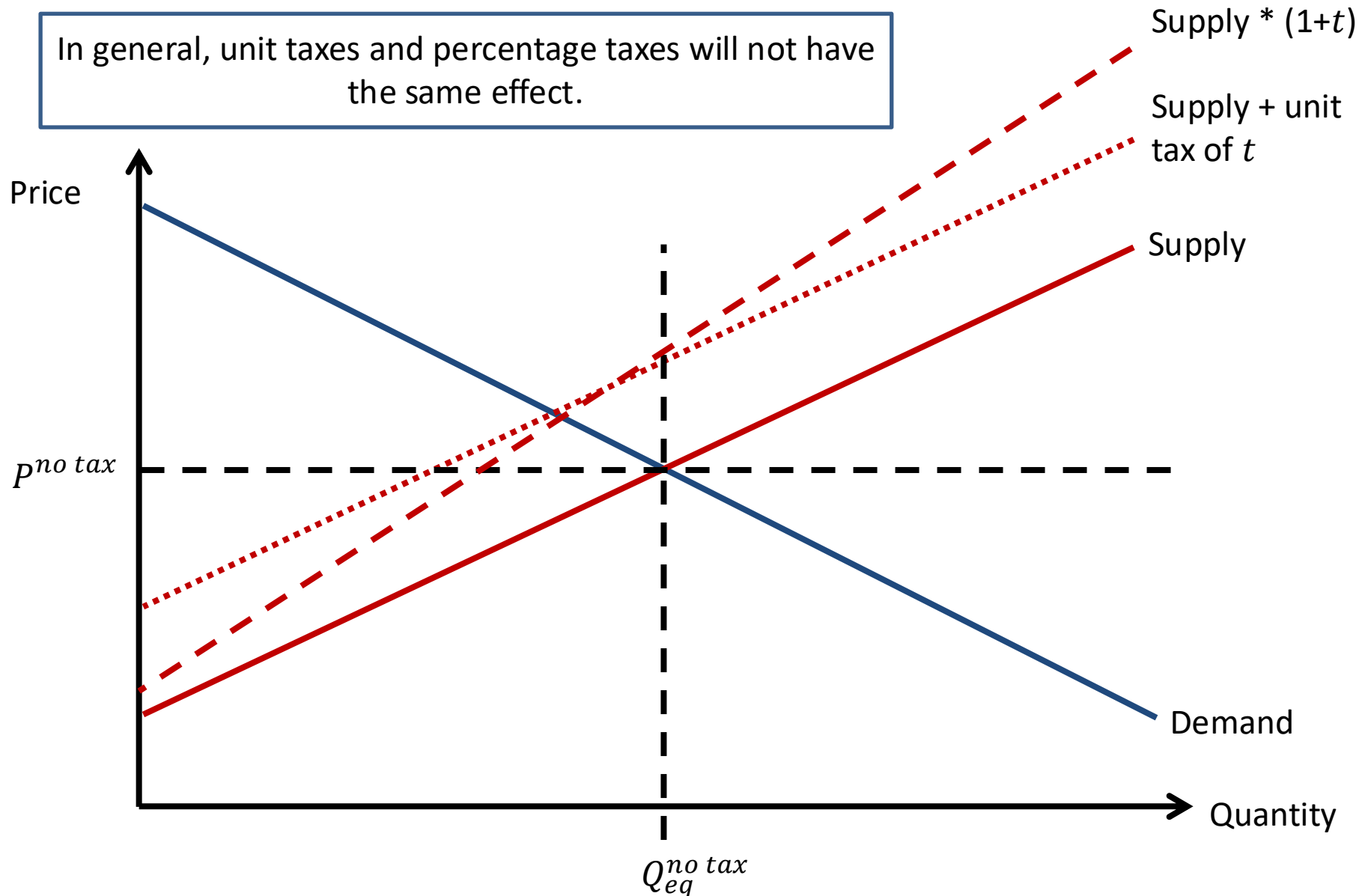
- Option C: Use the elasticity formula:

$$B_{cons} = \frac{\varepsilon_S}{\varepsilon_S + |\varepsilon_D|}, B_{prod} = \frac{|\varepsilon_D|}{\varepsilon_S + |\varepsilon_D|}$$

All three methods in Step 3 are equivalent and tell you who bears the burden (or “incidence”) of the tax.

Unit Tax vs Percentage Tax

In general, unit taxes and percentage taxes will not have the same effect.



Today's Plan

Government interventions:

1. Taxation

- How taxes work
- Dead weight loss
- Who pays



2. Application

- **Soda tax article**

3. Externalities

- Tax objectives
- Introduction to externalities
- “Corrective” taxation

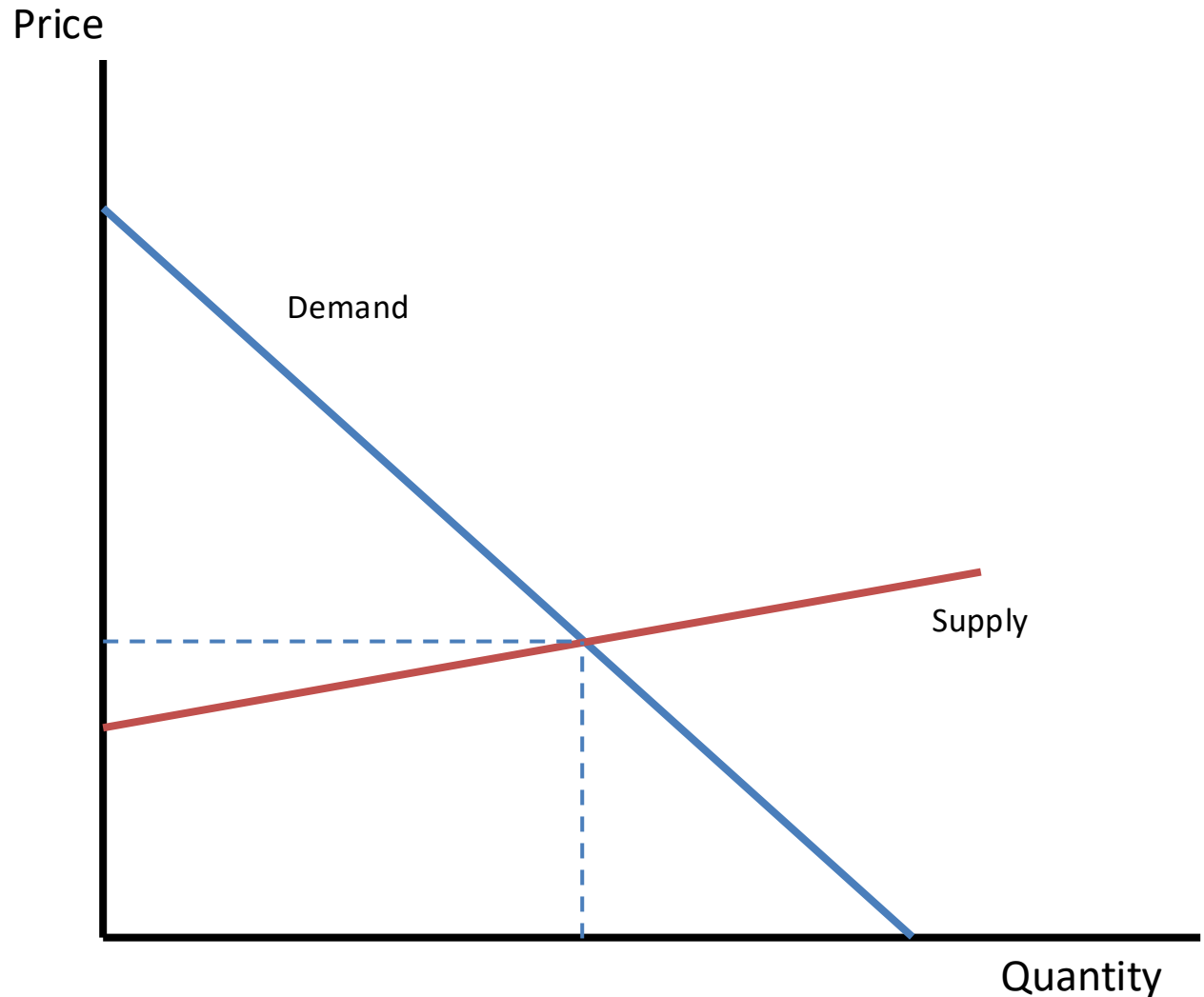
4. Other policy tools

- Quotas/rationing
- Price floors
- Cap and Trade

Application: Soda Tax Article

What does demand for sugary drinks look like?

What does supply look like?



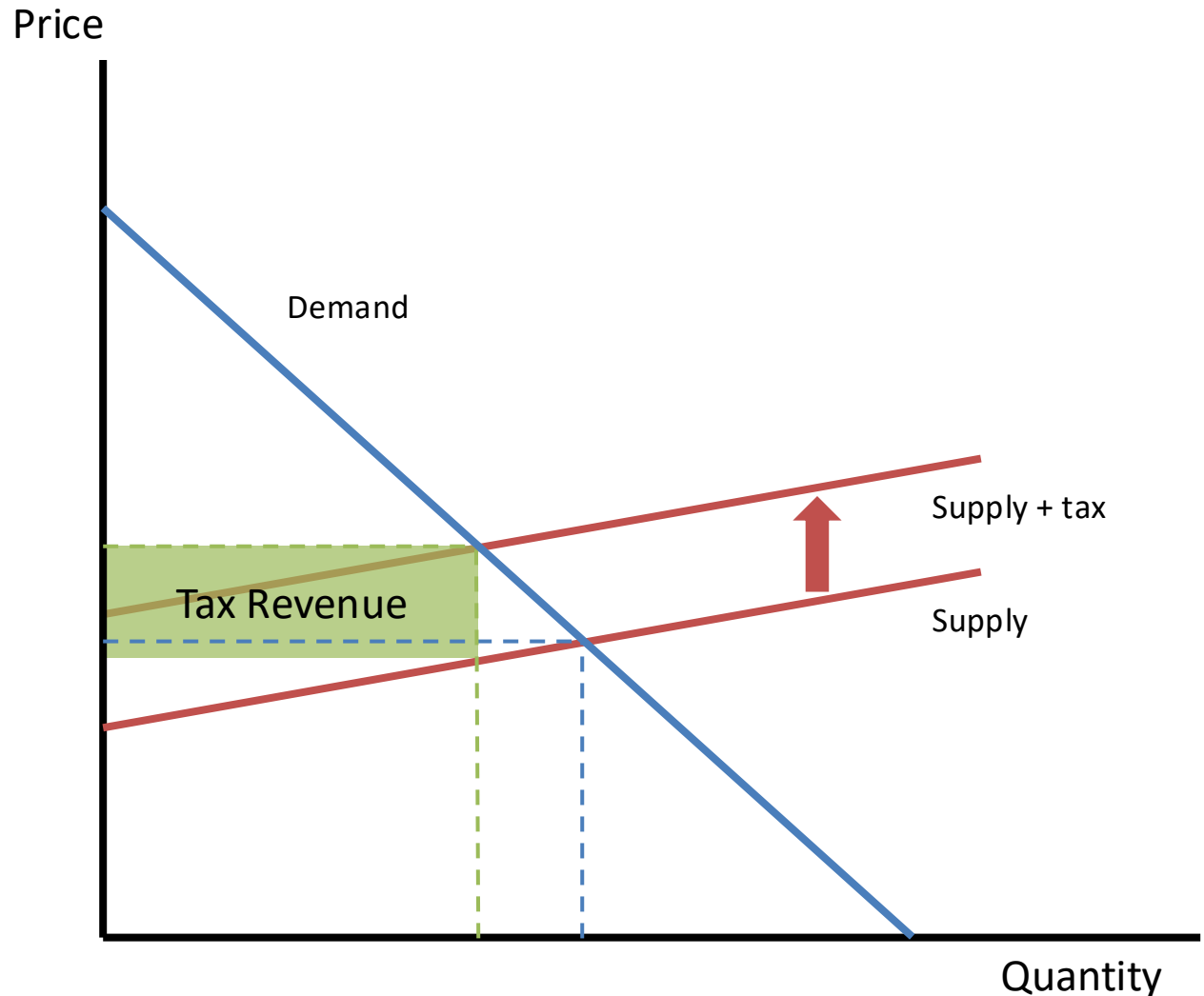
Application: Soda Tax Article

Suppose now we introduce a tax of 1.5 cents per oz:

How does a tax affect this market?

The new market equilibrium:

- Has a higher price
- Has a lower quantity.
- Results in new tax revenue.

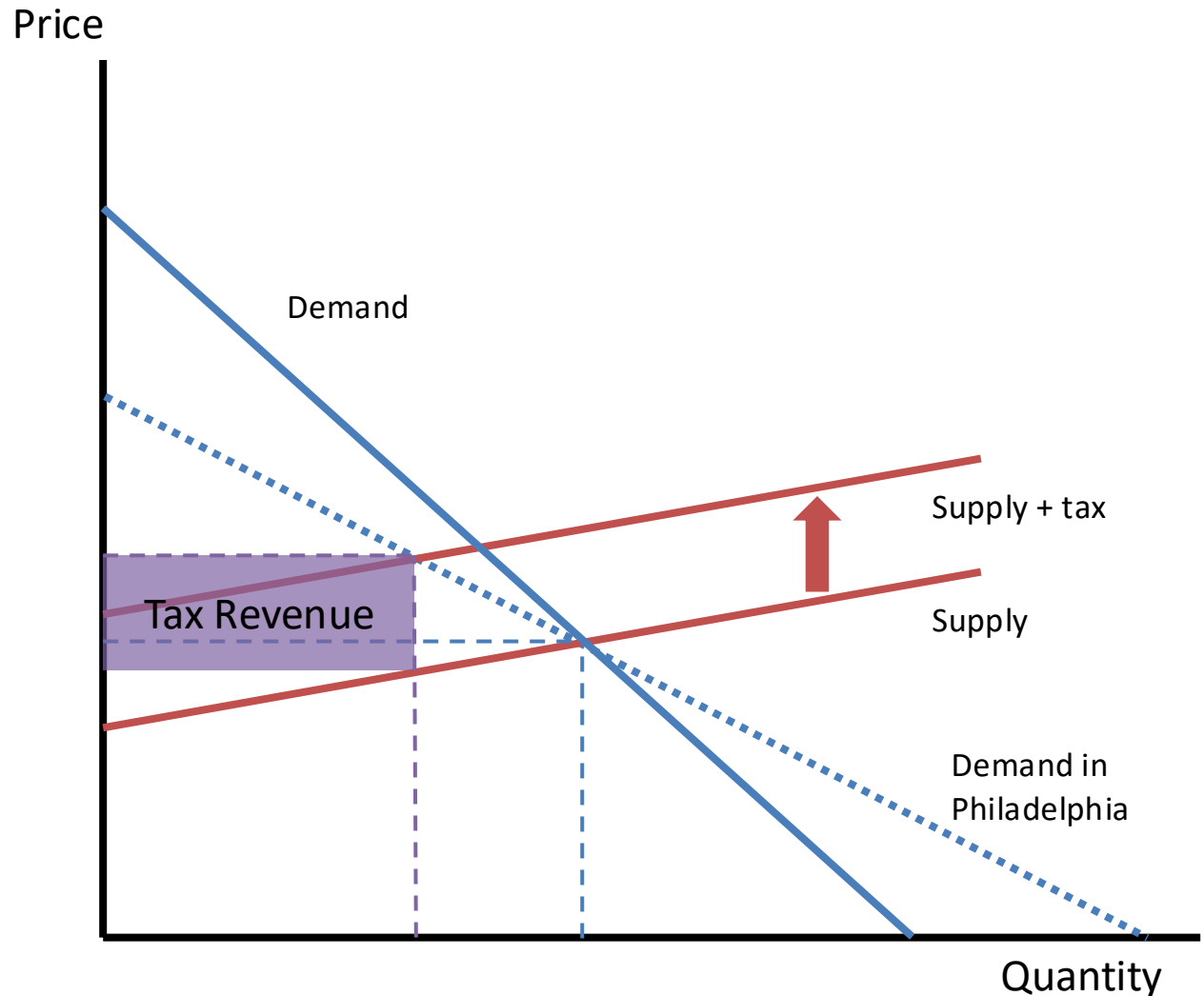


Application: Soda Tax Article

However, there are substitutes available to buying soda in Philadelphia.

- For example, buying soda in NJ, DE, or just drive 2 miles in any direction.

How does the presence of substitutes affect demand?



Cigarette Smuggling Flows and Tax Rates

2011 Consumption Smuggled (positive is inflow, negative is outflow)



2011 Cigarette Tax Rate (\$/pack)



Sources: Mackinac Center for Public Policy, Tax Foundation.

In the top map, rates are in percentage terms. Note: Smuggling rates are not available for Alaska, Hawaii, North Carolina and the District of Columbia. Cigarette tax rates have changed for some states since 2011. Map created using [Many Eyes](#).

Cross-border tax differences create real incentives:

- States with the highest cigarette taxes see the most cigarettes smuggled into the state.

Philadelphia Soda Tax

Many questions remain:

1. Are consumers harmed by this?
Which ones?
2. How do producers feel about this?
3. Is this a good way to raise revenue for the city's preschool program?



Economists are concerned about the *incidence* of taxes, and how much waste they may create.

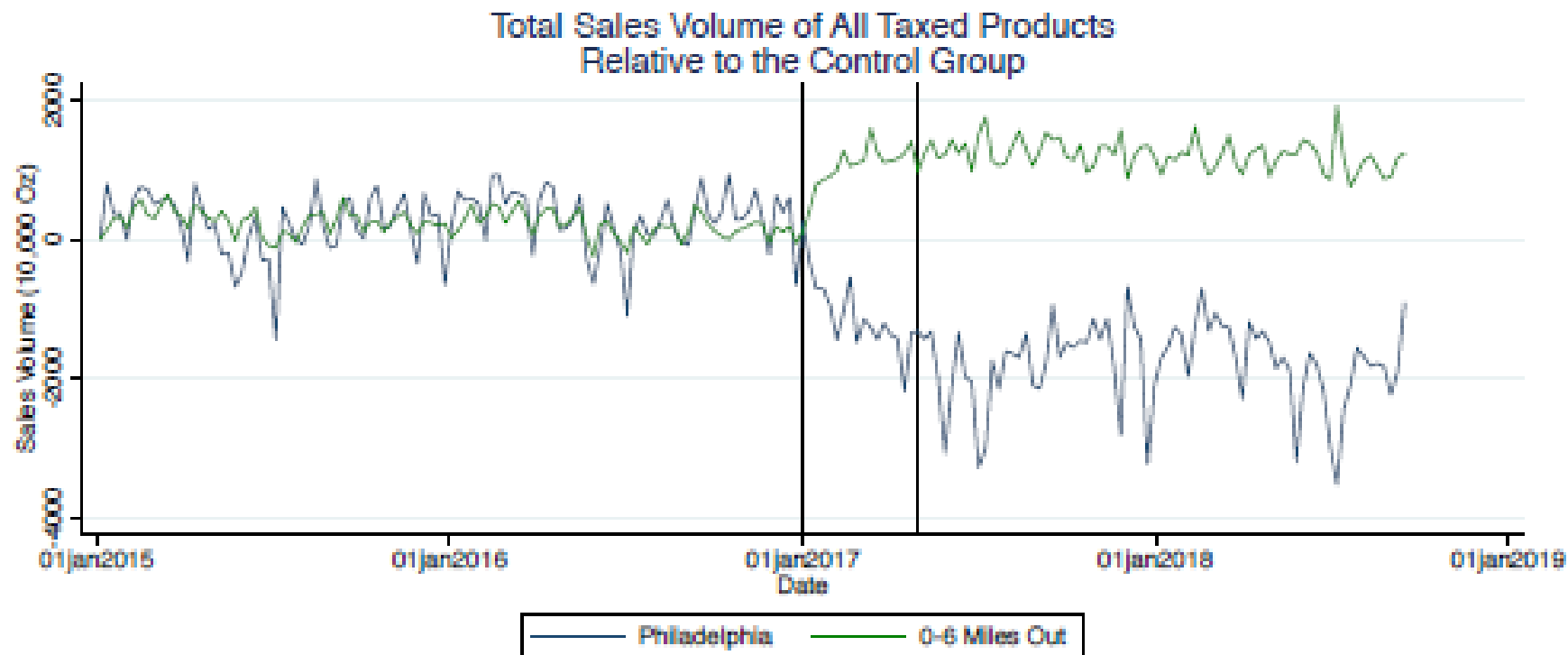
What was the objective of the city to justify creating that waste?

Preliminary evidence:

- Tax went into effect Jan 1, 2017. Prices have risen.
- Mayor Kenney: “They’re gouging their own customers.”
 - Media reports say that he thought that since it was “imposed” on wholesalers, consumers wouldn’t see prices rise.
 - Revenue is below expectations.
- Similar experiments: Berkeley, CA imposed a smaller surgery-drink tax.
 - Study* finds 43.1% of tax is passed-through (“paid”) by consumers.
 - Berkeley tax only on sugar. Philly tax includes artificial sweeteners. Which category has more inelastic demand? What does that predict?

*“THE INCIDENCE OF TAXES ON SUGAR-SWEETENED BEVERAGES: THE CASE OF BERKELEY, CALIFORNIA”, John Cawley and David Frisvold, NBER Working Paper

“Leakage” of Philly Soda Tax*



We find the tax is passed through at an average rate of 97%, leading to a 34% price increase... A large amount of cross-shopping to stores outside of Philadelphia offsets more than half of the reduction in sales in the city and decreases the net reduction in sales only 22%.

*The Impact of Soda Taxes: Pass-Through, Tax Avoidance, and Nutritional Effects, Journal of Marketing Research, 2021, 58(1), 22-49,

An Alternative Approach: UK Sugar Tax

Officials in the UK took a different approach, with two tiers of sugar taxes:

Beverages with...

Between 5 and 8g of sugar
per 100ml

More than 8g of sugar per
100ml

... face a tax of...

- 18p per liter
 - Equivalent to 24 cents per liter
- 24p per liter
 - Equivalent to 32 cents per liter



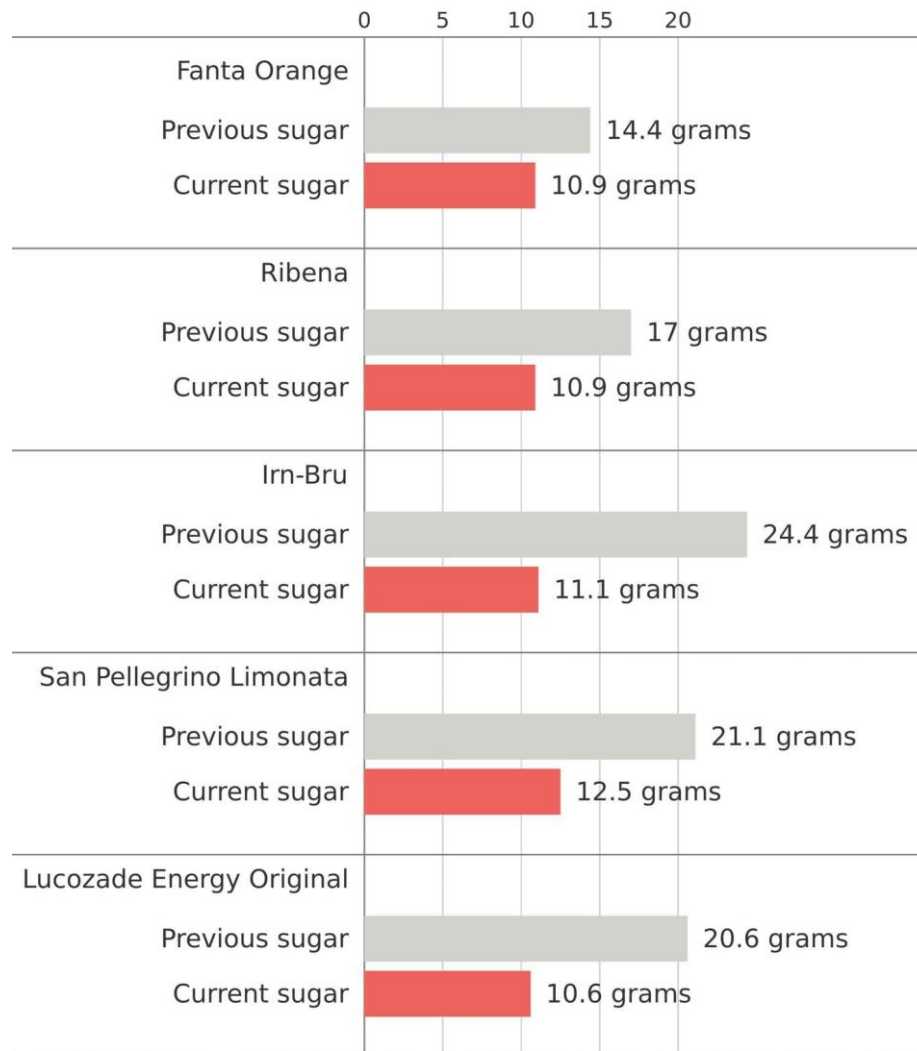
How might the impact of this tax be different than the Philadelphia tax?
(Philadelphia's tax works out to almost 51 cents per liter)

Same: both will be passed (at least partially) on to consumers, who should decrease consumption.
Different?

The British Tax Design Creates New Incentives

British soda-makers are slashing sugar

Some brands have changed their recipes to avoid a new tax, which applies to soft drinks with more than roughly 12 grams of sugar per eight-ounce can.



The British sugar tax creates additional incentives:

- If you can get your product under the 8g threshold, or under the 5g threshold, your customers effectively get a price cut
- If the goal is to reduce sugar intake, you get both the benefit from some consumers reducing consumption, AND the benefit of those that continue to consume having a lower intake.
- “Equilibrium” here involves suppliers changing their products, unlike in Philadelphia.

Cost of Taxation

Imposing taxes costs society in two ways:

Deadweight Loss

- The tax distorts the market. A tax can prevent some customers from buying a good that they would otherwise buy. (Exception: if demand or supply is perfectly inelastic there is no deadweight loss)

Administrative Cost

- It costs money to collect and regulate the tax. In theory this should be small. In practice?



Taxes are a necessary evil in the economy, but they should be designed as simply as possible to minimize administrative costs.

Administrative Costs: New York's Sandwich Tax



PUBLICATIONS

Publications by number

General information publications

Publications by tax type

Technical documents

Regulations and regulatory actions

New York State tax law

Recent additions

NYS Division of Tax Appeals

IRS publications

Sandwiches

Tax Bulletin ST-835 (TB-ST-835)

Issue Date: April 13, 2011

Introduction

Sandwiches are generally subject to sales tax. This bulletin explains what is considered a sandwich for s listed at the end of this bulletin.

What is considered a sandwich

Sandwiches include cold and hot sandwiches of every kind that are prepared and ready to be eaten, whet or as elaborate as a six-foot, toasted submarine sandwich.

Some examples of taxable sandwiches include:

- common sandwiches, such as:
 - BLTs (bacon, lettuce, and tomato sandwiches);
 - club sandwiches;
 - cold cut sandwiches;
 - grilled cheese sandwiches;
 - peanut butter and jelly sandwiches
 - salad-type sandwiches (e.g., chicken, egg, ham, and tuna);
- bagel sandwiches (served buttered or with spreads, or otherwise as a sandwich);
- burritos
- cheese-steak sandwiches;
- croissant sandwiches;

New York's Dept. of Taxation released a tax bulletin on "what is considered a sandwich"

- Hot dog?
 - Sandwich.
- Buttered bagel?
 - Sandwich.
- Burrito?
 - Sandwich.
- Taco?
 - ... unclear!

Administrative Costs: Jaffa Cakes



UK's VAT is payable on “chocolate-covered biscuits”, but not on “chocolate-covered cakes”.

- The manufacturer wanted to claim they were cakes and not biscuits, even though they were shaped and sold like biscuits.
- There was a trial in the 90s.
- The clincher: “It goes hard like a cake when stale in oppose to soft like a biscuit would.”

Simple taxes can lead to complications and create odd incentives.

Administrative Costs: The Snuggie



An existential question: is the Snuggie a blanket or a garment?

- Import tariffs for garments is 14.9%, for blankets is 8.5%

Slip Op.17-15

UNITED STATES COURT OF INTERNATIONAL TRADE

ALLSTAR MARKETING GROUP, LLC,

Plaintiff,

v.

UNITED STATES,

Defendant.

Before: Mark A. Barnett, Judge

Court No. 13-00395

OPINION

[The court finds that the subject imports are properly classified as "blankets" under HTSUS subheading 6301.40.00. Accordingly, the court grants Plaintiff's motion for summary judgment and denies Defendant's cross-motion for summary judgment.]

Dated: February 10, 2017

Joseph M. Srinan, Robert B. Silverman, and Frank J. Desiderio, Grinfeld

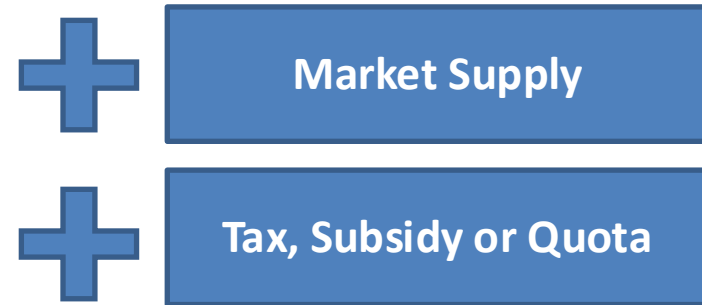
Verdict: blanket.

Market Interventions



Market Equilibrium

- Consumer surplus
- Producer surplus



New Market Equilibrium

- Intervention changes consumer surplus, producer surplus and deadweight loss
- Elasticities tell us how

Why do governments intervene in markets?

Governments require revenues to operate. However, one way in which government interventions can benefit society is to **improve equilibrium outcomes in the presence of externalities**. Other reasons?

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Motivation: Externalities

The situation:

Many production processes create pollution that harms the environment and has adverse effects on human health and the planet

- Coal plants produce CO₂, SO₂, NO_x, that can cause smog, acid rain, climate change
- Aluminum plants produce toxic sludge that can escape from its ponds (Hungary)
- What else?



The problem:

The cost of these pollutants is not borne by the producers, it is borne by those affected...

- Local residents who drink the water and breathe the air
- In the case of climate change, this is *everyone*



Externalities

Externalities

An externality occurs when a person's well-being or a firm's production capability is directly affected by the actions of other consumers or firms (rather than indirectly through changes in prices).

Externalities can be...

Positive

Makes others *better off*

Negative

Makes others *worse off*

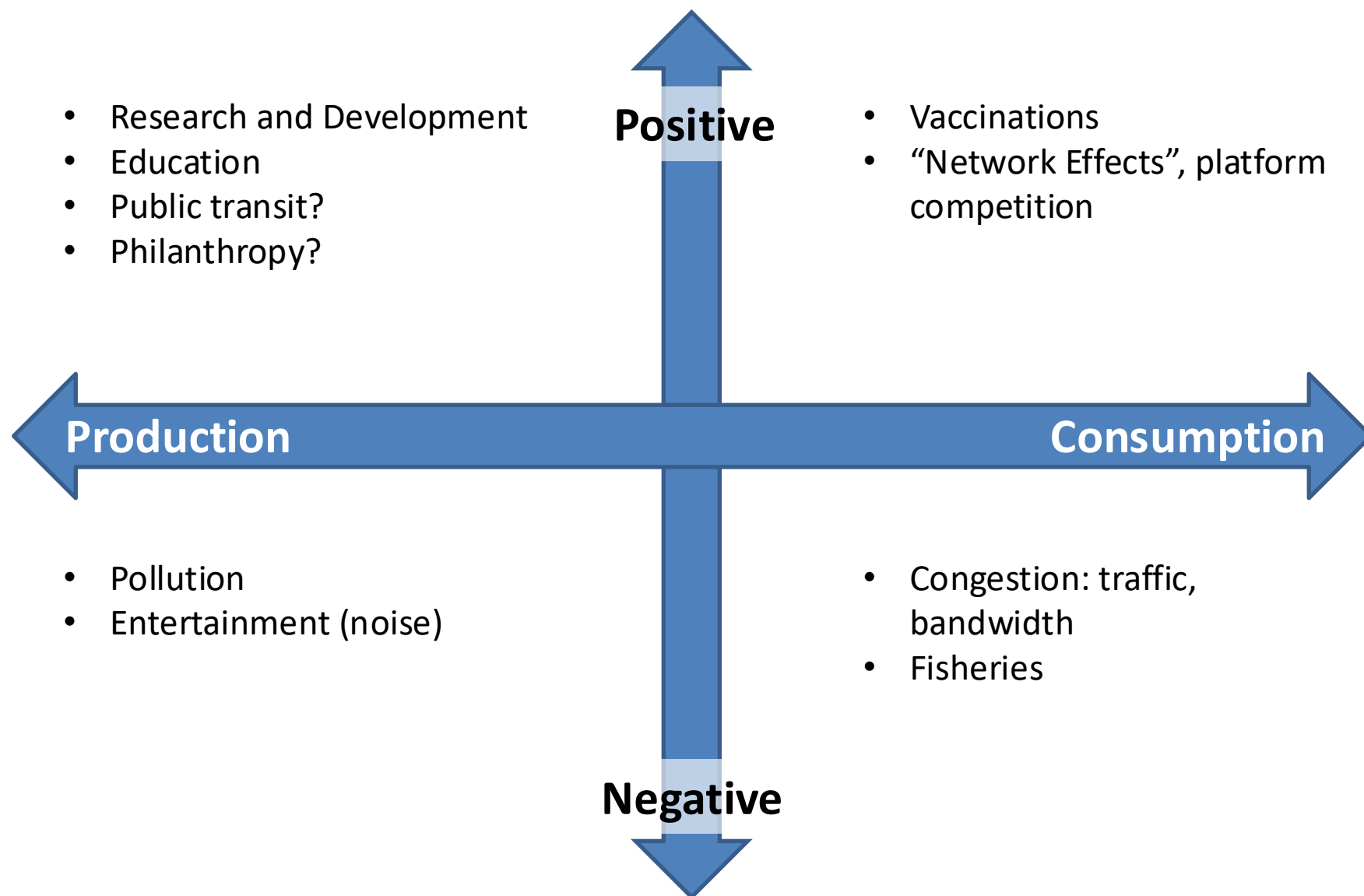
In Production

The benefit/harm arises when a good is *produced*...

In Consumption

The benefit/harm arises when a good is *consumed*...

Examples of Externalities



The Problem

Over- (and Under-) production

Since the decision maker does not incur the **full** cost or benefit of the consumption/production that leads to the externality, the decision maker does not produce the socially optimal amount

In a competitive market, goods with negative externalities are over-produced and goods with positive externalities are under-produced...

Cost of producing an additional unit

Private cost faced by the firm (MC)

External cost of production (MEC)

- *Marginal External Cost*

Marginal social cost $MSC = MC + MEC$

Benefits

Private benefit of the firm ($MR=P=MB$)

External benefit to consumption (MEB)

- *Marginal External Benefit*

Marginal social benefit $MSB = MB + MEB$

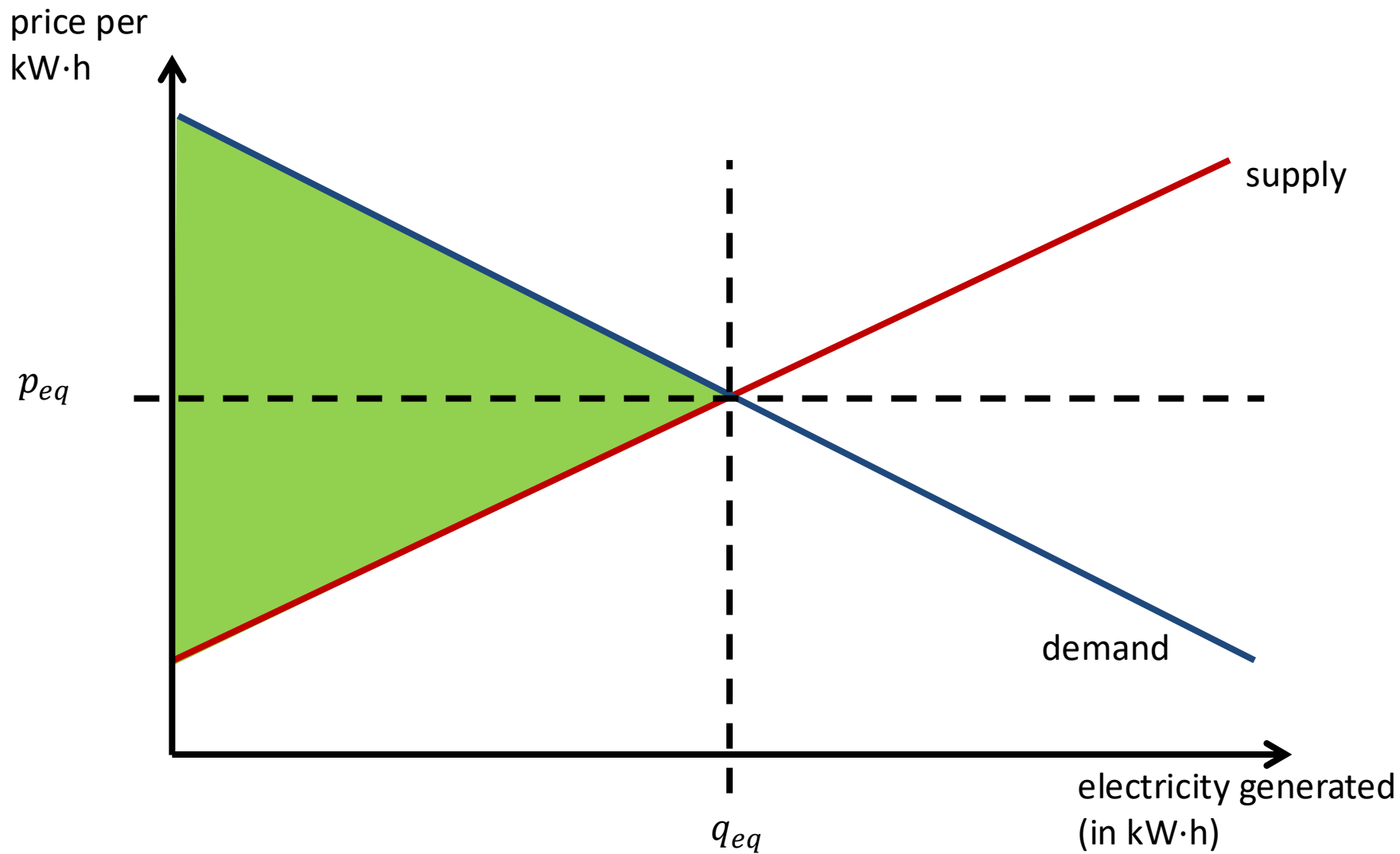
What maximizes profit for the firm?

$MC=MB$

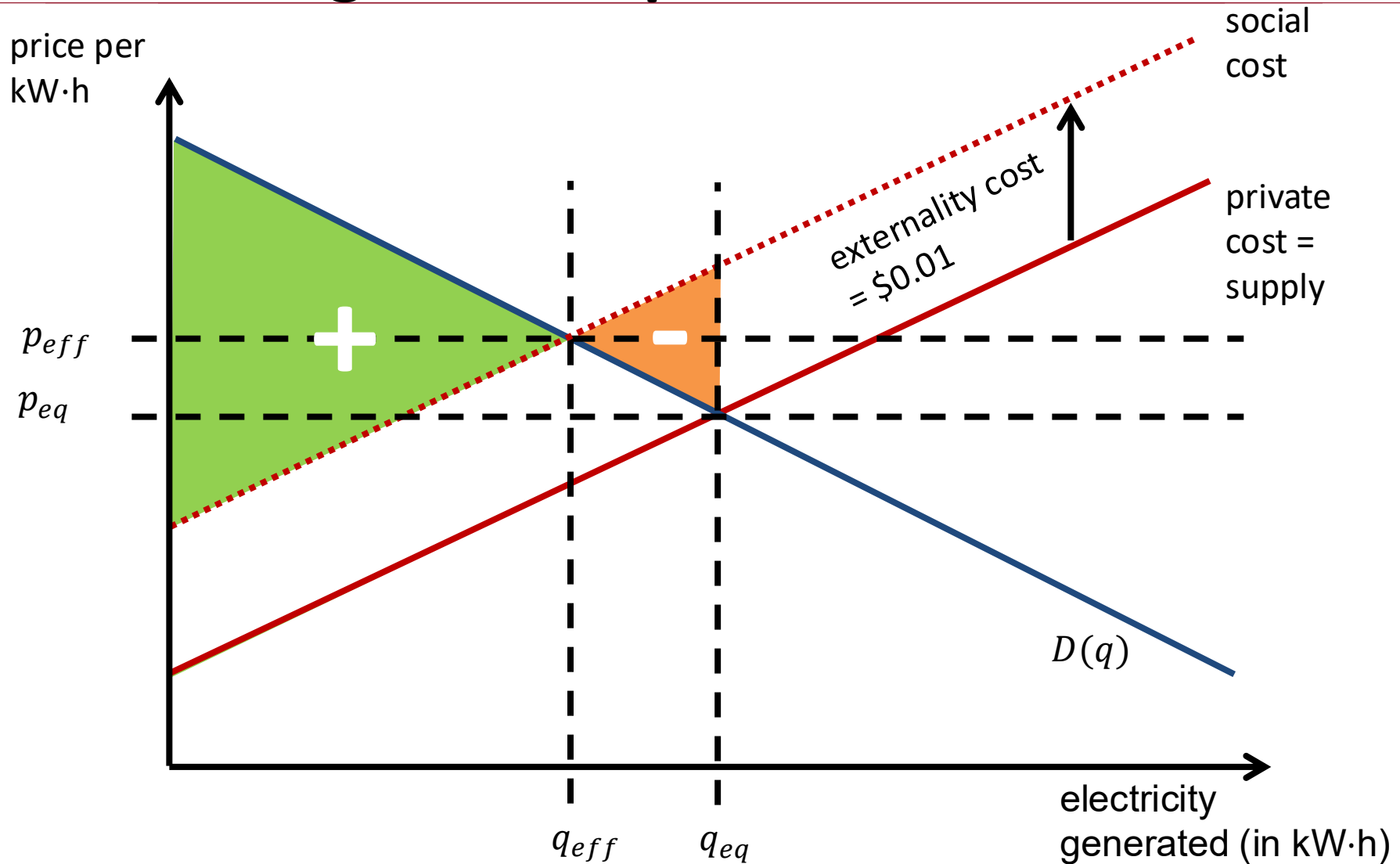
What maximizes social welfare?

$MSC=MSB$

Overcounting total surplus



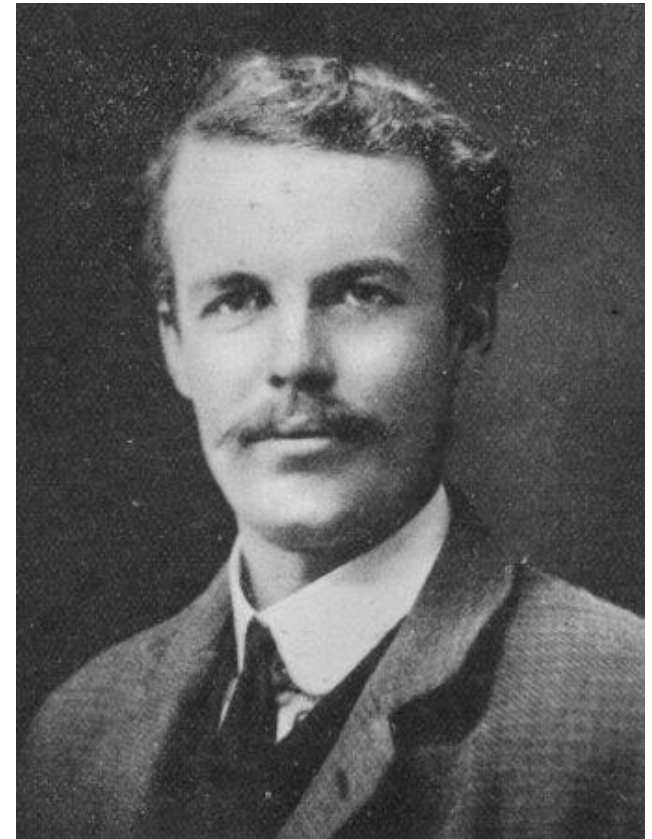
Overcounting total surplus



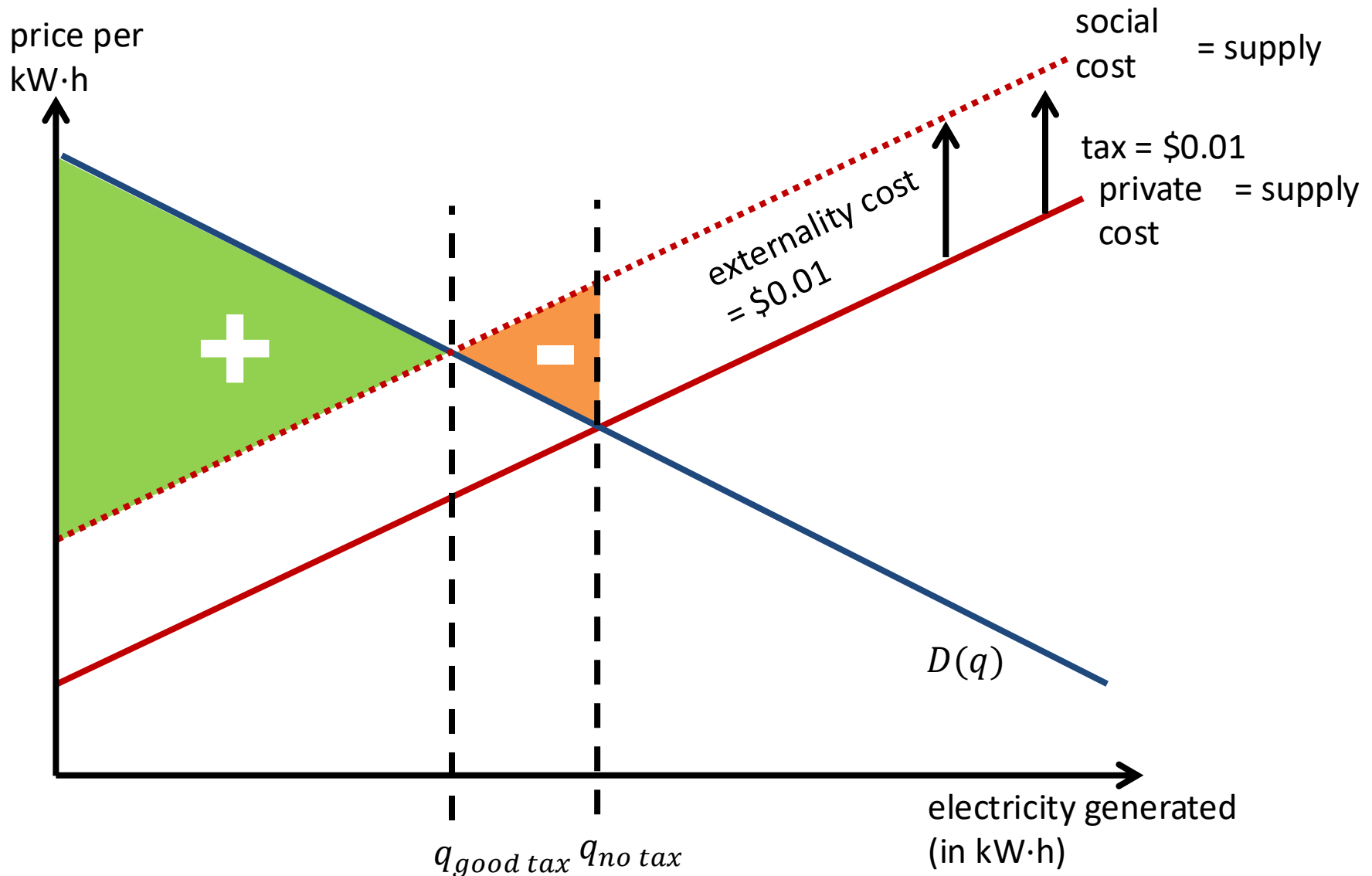
“Corrective” taxation: Pigou

Making a tax equal to the social cost effectively “internalizes the externality”.

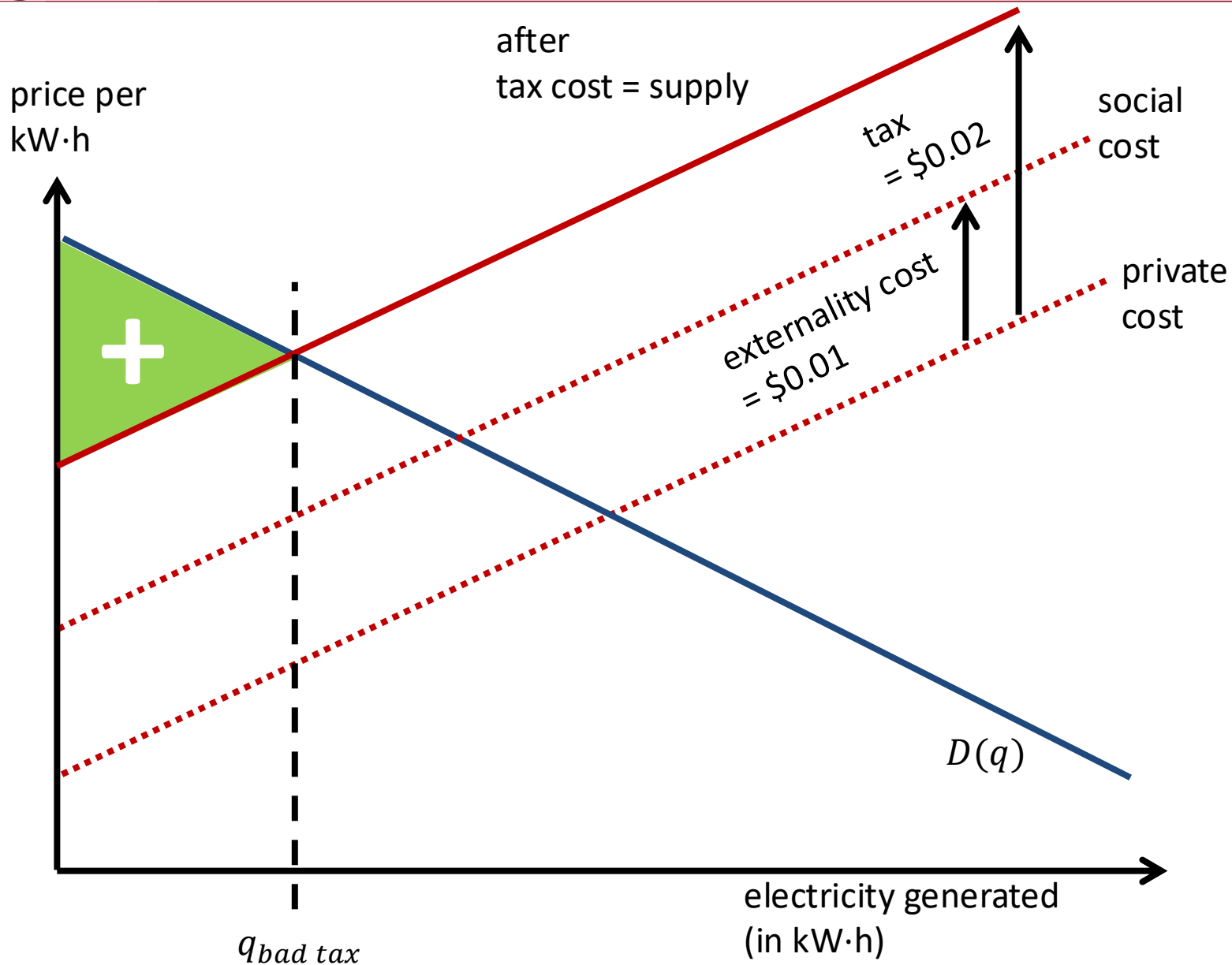
i.e., the parties to a bilateral transaction take into account the harm their transaction will impose on everyone else. Of course, making the tax too big might make matters worse!



Pigouvian taxation: before and after



Pigouvian taxation: too much



Today's Plan

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- How taxes work
- Dead weight loss
- Who pays (elasticities matter!)

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- Tax objectives
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- “Corrective” taxation

4. Other policy tools

- **Quotas/rationing**
- **Price floors**
- **Cap and Trade**

Other Approaches besides Taxes

Price Restrictions

- Sets a range of prices that are allowable.
- Will you be at an equilibrium? Maybe...

Quotas

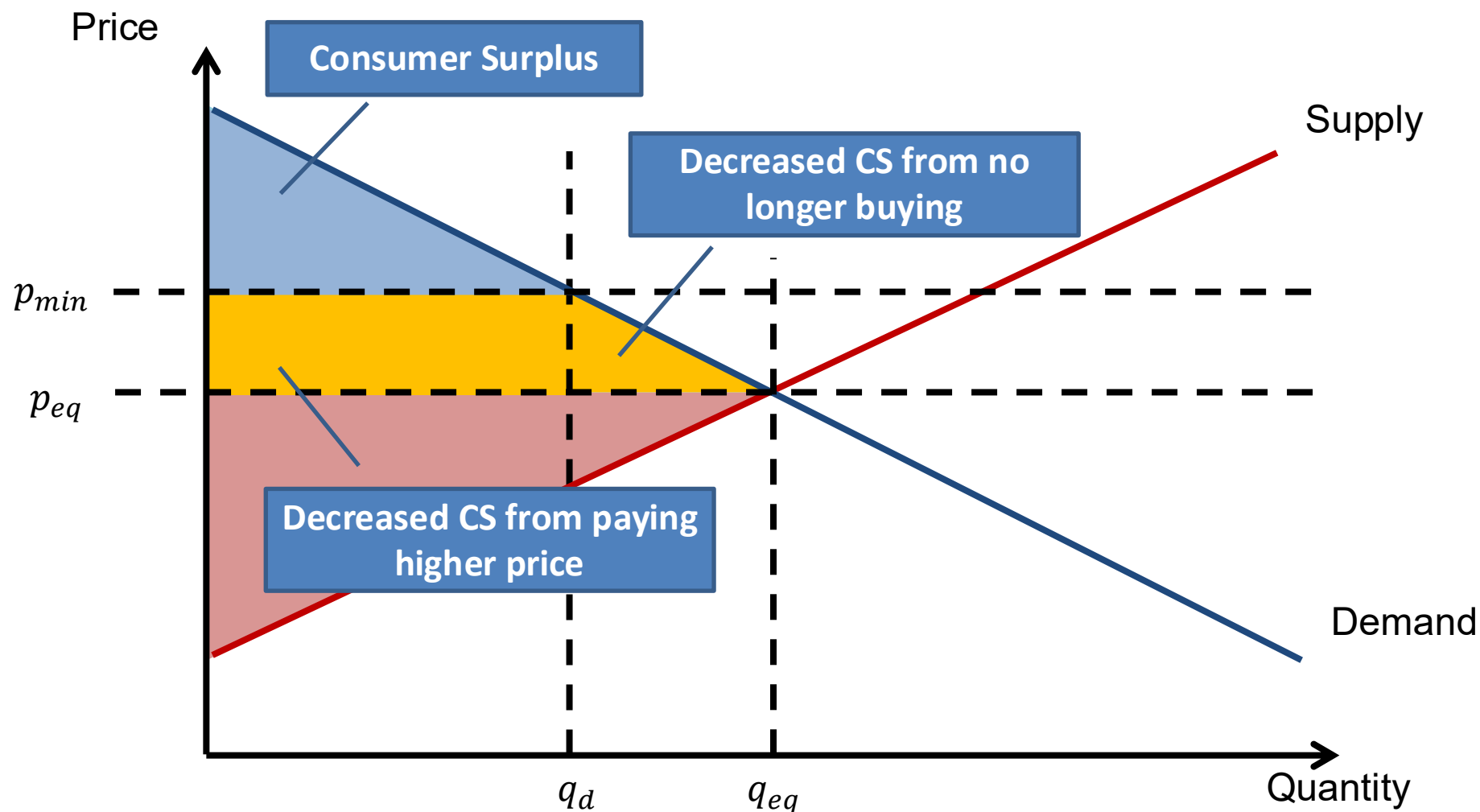
- Sets a range of quantities that are allowable.
- Will you be at an equilibrium? Maybe...



These can be difficult to implement. What is the right price floor or ceiling? What is the right quota for each firm to produce or each individual to consume?

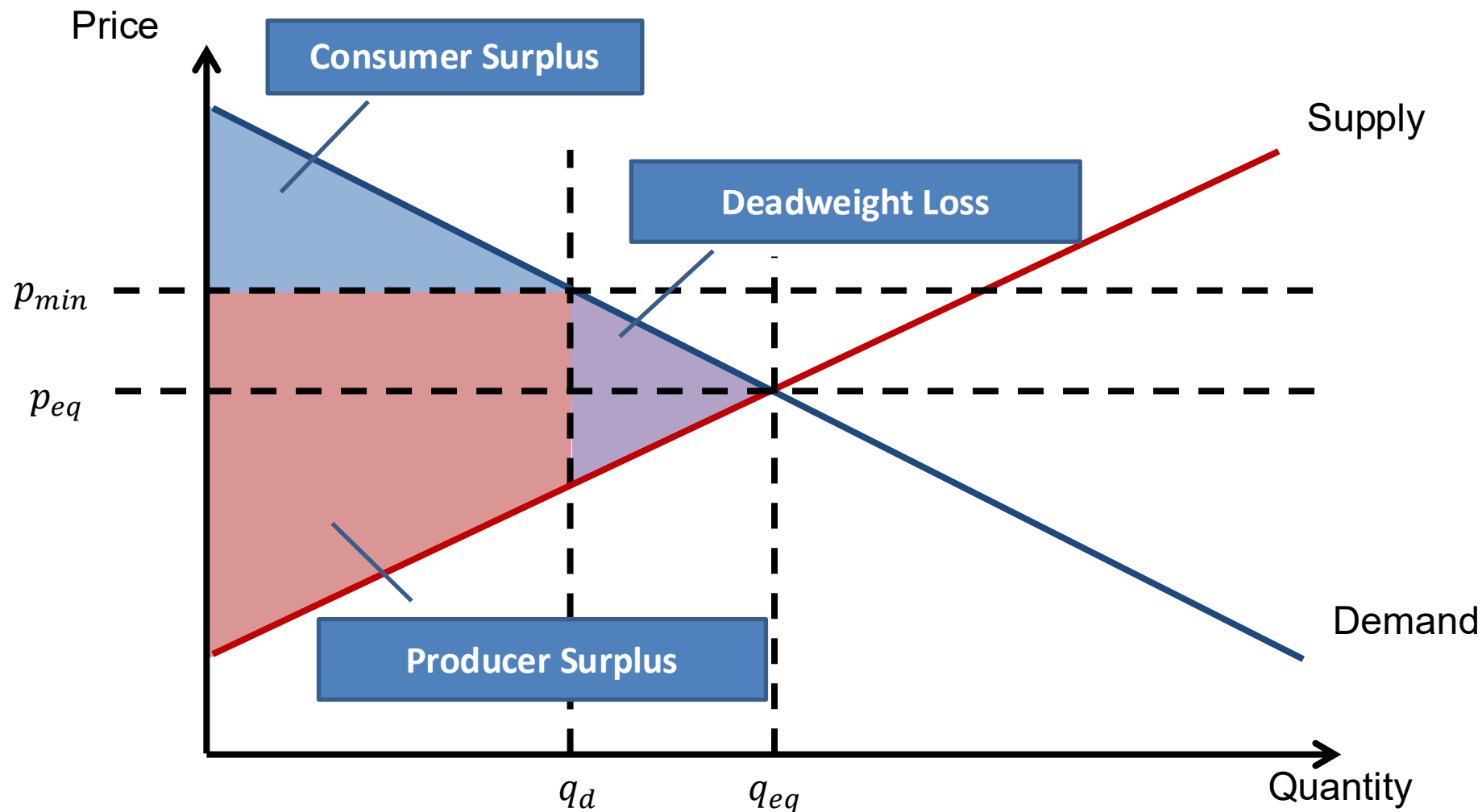
These are “blunt instruments” to intervene in a market. Very easy to be “out of equilibrium”.

Introduce a Binding Minimum Price



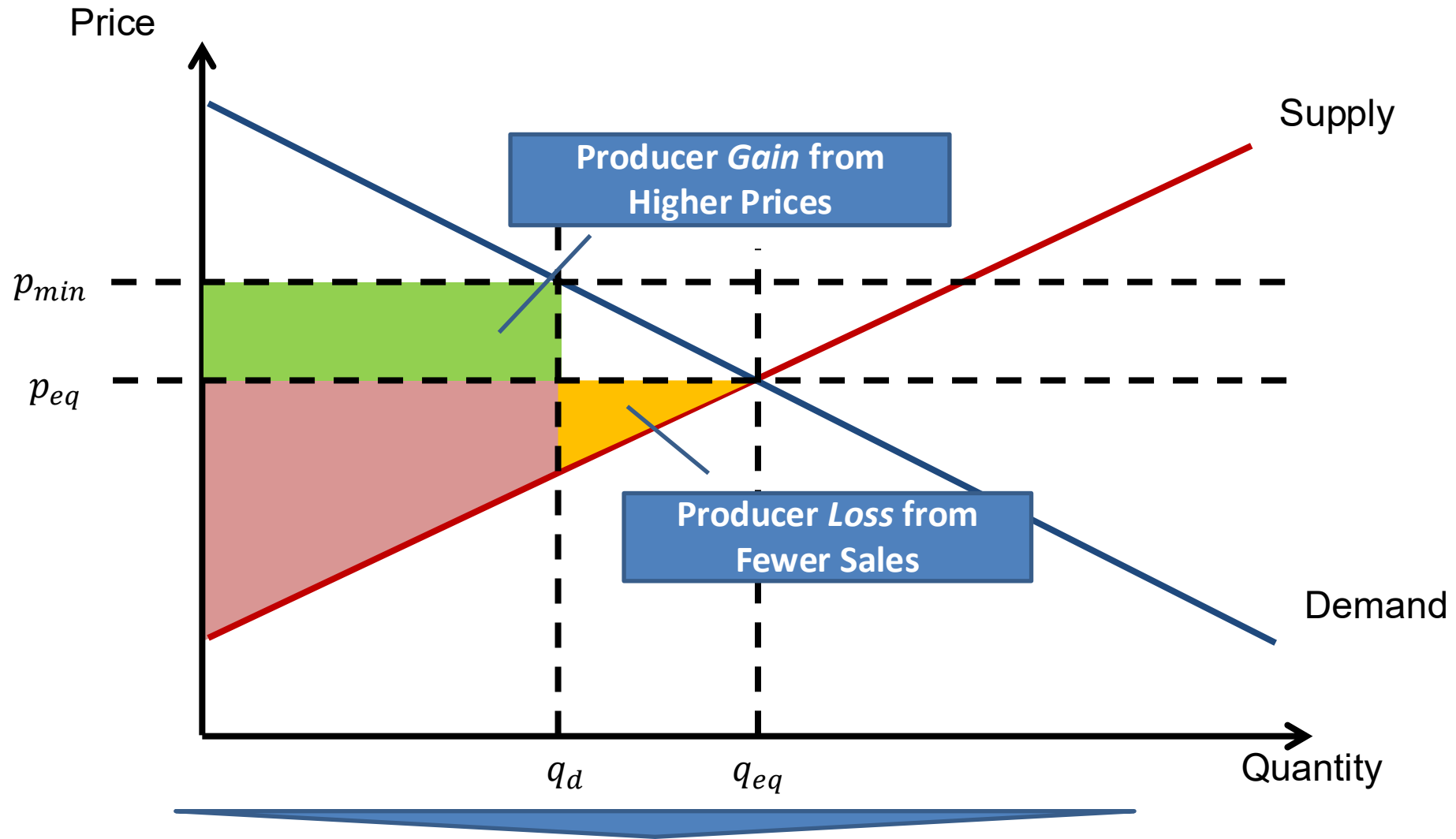
A binding minimum price reduces Consumer Surplus

Minimum Price: Market Outcome



Creating DWL in order to transfer to producers

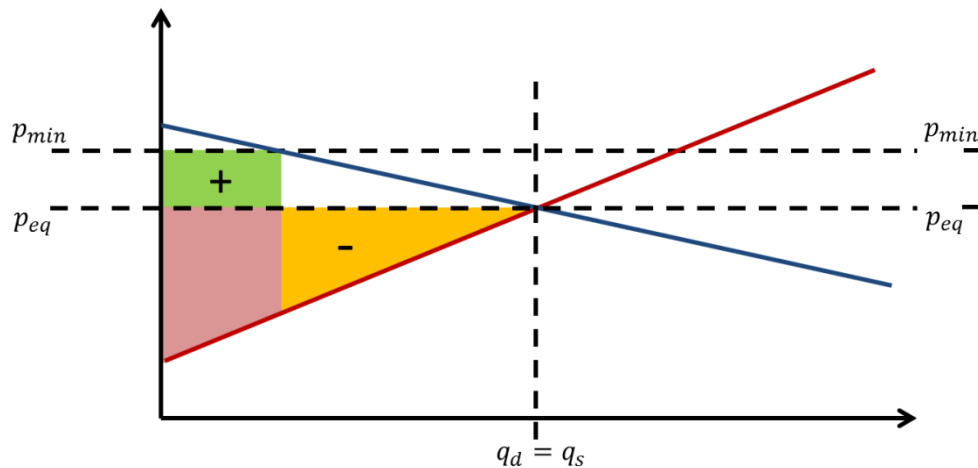
Minimum Price: Producer Benefit



Producers would only favor this if the gain outweighs the loss.
When is that likely to occur?

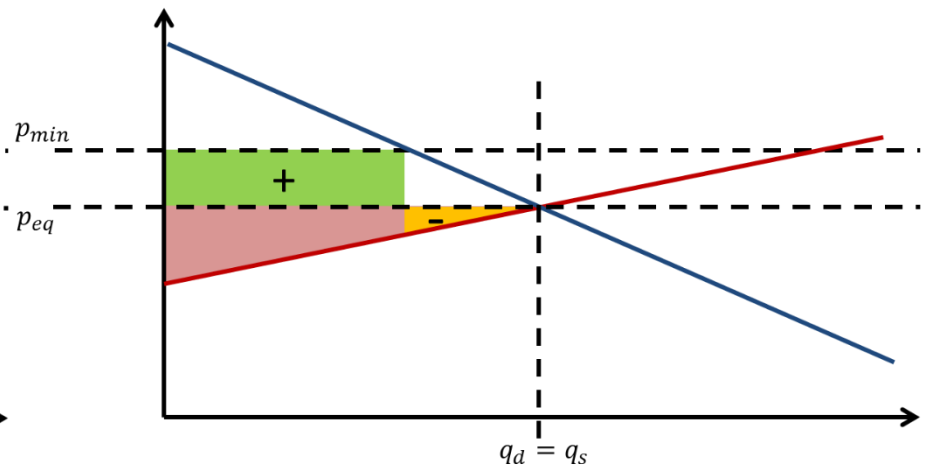
When Would Producers Favor This?

More elastic demand?



Decreases producer gain, increases producer loss. So, LESS likely for producers to favor price controls.

More elastic supply?



Decreases loss from reduction in quantity. So, MORE likely for producers to favor price controls.

Producers and Minimum Prices

Producers are more likely to favor minimum price laws when demand is inelastic, and supply is elastic. Examples?

How does this apply to the minimum wage?

Rationing as a Policy Tool

World War I: USA avoided rationing; used voluntary programs instead

Hang this where you will see it every day.

United States Food Administration



REMEMBER THE DAYS

SUNDAY—One meal Wheatless; one meal Meatless.
MONDAY—All meals Wheatless; one meal Meatless.
TUESDAY—All meals Meatless; one meal Wheatless.
WEDNESDAY—All meals Wheatless; one meal Meatless.
THURSDAY—One meal Wheatless; one meal Meatless.
FRIDAY—One meal Wheatless; one meal Meatless.
SATURDAY—All meals Porkless; one meal Wheatless; one meal Meatless.

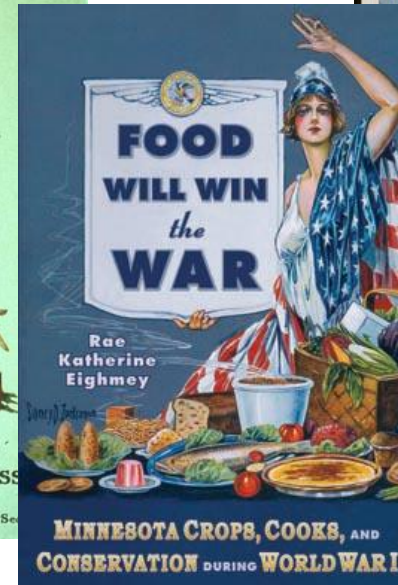
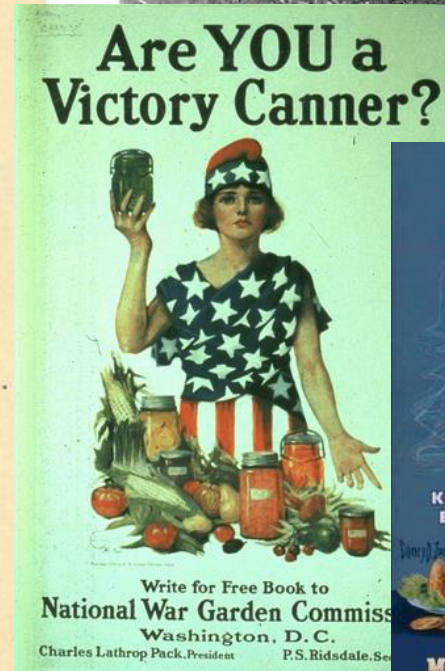
On Wheatless Days
 Use no bread, crackers, pastry or breakfast foods containing wheat. Use no wheat flour in cooking except the smallest amount necessary for thickening soups and gravies, or as a binder in corn or other cereal breads. If you buy bread, pastry or cakes from baker or grocer, buy only wheatless products or Victory Bread (containing at least one-fifth other cereals than wheat) or Victory Pastry, etc. (containing at least one-third other cereals than wheat).

On Meatless Days
 Use no beef, pork or sheep products.

On Porkless Days
 Use no pork, bacon, ham, lard or other pork products, fresh or preserved.

On All Days
 With each purchase of wheat flour for home use you must buy also an equal amount of other cereals (corn meal, corn starch, corn flour, hominy, corn grits, barley flour, rice, rice flour, oatmeal, rolled oats, buckwheat flour, potato flour, sweet potato flour, etc.) The housewife or cook may use these non-wheat products separately in making bread, cakes and pastry, or may mix them with wheat flour, except for wheatless meals.

February, 1918.
 Index No. F. 4.



Rationing as a Policy Tool

World War II: direct consumer rationing: rubber, gasoline, metals, food.

UNITED STATES OF AMERICA
OFFICE OF PRICE ADMINISTRATION

No. 920315 BM

WAR RATION BOOK No. 3 Fold if altered

Identification of person to whom issued: PRINT IN FULL

(First name) (Middle name) (Last name)

Street number or rural route _____

City or post office _____ State _____

AGE	SEX	WEIGHT Lbs.	HEIGHT Ft. In.	OCCUPATION

SIGNATURE _____
(Person to whom book is issued. If such person is unable to sign because of age or incapacity, another may sign in his behalf.)

WARNING
This book is the property of the United States Government. It is unlawful to sell it to any other person, or to use it or permit anyone else to use it, except to obtain rationed goods in accordance with regulations of the Office of Price Administration. Any person who finds a lost War Ration Book must return it to the War Price and Rationing Board which issued it. Persons who violate rationing regulations are subject to \$10,000 fine or imprisonment, or both.

LOCAL BOARD ACTION

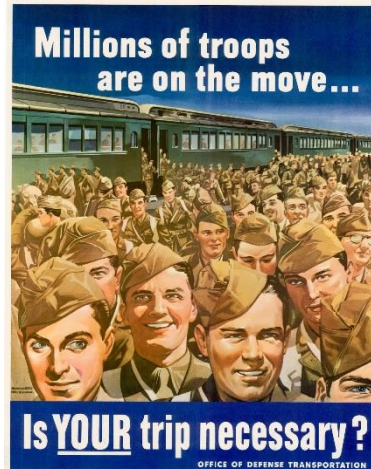
Issued by _____ (Local board number) _____ (Date) _____

Street address _____

City _____ State _____

(Signature of issuing officer)

OPA Form No. R-130



Need to think about what goods are complements, substitutes as well

Cap and Trade

How do we generate the socially optimal amount of power production

Tactic:	Strategy	Concerns
Quotas (“Emissions Standards”)	• Set a maximum output amount for each power plant	• Requires knowledge of optimal amount for each power plant to produce
Taxes (“Emissions Fees”)	• Tax the creation of pollution	• What rate? Depends on current technology, etc. May change over time.
Cap and Trade	• Cap the total amount of pollution and sell permits with a secondary market.	• What amount? Cost of running the system? Incentives for firms?

Some firms can decrease the external cost (“abate”) more easily and inexpensively than others. The ideal system would incentivize these firms to take those abatement measures

Article: “Cap and Trade”

Example: What if firms experience differential abatement costs? (in millions?)

Big Power 1		Big Power 2	
Emissions	Cost	Emissions	Cost
4	\$0	4	\$0
3	\$1	3	\$3
2	\$2	2	\$6
1	\$3	1	\$9
0	\$4	0	\$12

Currently both emit 4...
but the government will set the total cap at 6.

What will happen if both firms are given a quota of 3?

Total abatement costs: \$4mil

Article: “Cap and Trade”

Example: What if firms experience differential abatement costs? (in millions?)

Big Power 1		Big Power 2	
Emissions	Cost	Emissions	Cost
4	\$0	4	\$0
3	\$1	3	\$3
2	\$2	2	\$6
1	\$3	1	\$9
0	\$4	0	\$12

- Lower abatement costs and incentive to invest in clean technology, because you get paid for it.
- Government cuts quotas over time as firms invest

Can we do better than the \$4mil abatement cost?

What will happen with cap and trade (still initial endowment of 3 each)?

Total abatement costs: \$2mil

Also: BP1 receives a payment from BP2 (value \$2-\$3mil)

Carbon Market

CARBON PRICE

\$/Tonne CO₂e



Price of California Carbon Allowance Futures over time from [ICE End of Day Reports](#).
Daily trading volume units are 1000 allowance futures. [Download source data](#).

California carbon market:

- Prices stable(ish) since 2013 start of trading
 - In 2015, expanded to cover transportation: estimates are \$0.11 per gallon increase in gas prices.
 - Generated over \$500M in 2013 for state revenue.
 - Much less in 2015, as companies invested in carbon reduction tech.
- Everything is more elastic in the long run.**

Indirect effects

- Predictable policy and market encourage innovation and new business opportunities. California hopes to link up other jurisdictions to their marketplace for carbon.



The result?



Carbon capture at Wisconsin farms to sell credits into California's cap-and-trade market.

Summary and Direction

Today

How do policy interventions in markets affect outcomes?

- Tax Incidence: depends on elasticity of supply/demand. Who “remits” doesn’t matter.
- If a tax is imposed on a good, the inelastic side of the market pays.
- Externalities: my choices don’t take into account the full cost/benefit to society.
- Taxes create a deadweight loss, but can also be used to correct for externalities.
- Cap and trade can be an efficient way to address externalities, but you need to know the level.

Next Lecture

Production

- Understanding firm costs
- Finding cost-minimizing mix of production inputs
- Revisiting where supply curves comes from
- Changing production in response to shocks